Tianyu Kong

kong0226@umn.edu   timkong98.github.io   timkong98

Research Interests

Mathematical modeling, numerical analysis, partial differential equations, multiple-scale analysis, mathematical physics, quantum physics, and optimization

Education

University of Minnesota Twin Cities, Minneapolis, MN — *09.2021 – 2026 (Expected)*
PhD in Applied Mathematics

- Advisor: Mitchell Luskin, Alexander B. Watson
- GPA: 3.96/4.00

University of Chicago, Chicago, IL — *09.2017 – 06.2021*
Honors BS in Applied Mathematics & Physics

- GPA: 3.73/4.00

Honors and Scholarships

UMN Doctoral Dissertation Fellowship	*05.2024*
SIAM Student Travel Award	*03.2024*
UMN Vanky Men Memorial Fellowship	*12.2023*
UChicago Dean's List	*07.2018, 07.2019*

Publications

1. **Interacting Twisted Bilayer Graphene with Systematic Modeling of Structural Relaxation**
   Tianyu Kong, Alexander B. Watson, Mitchell Luskin, Kevin D. Stubbs
   Electronic Structure. 035001 (2025)
2. **Higher-order continuum models for twisted bilayer graphene**
   Solomon Quinn, **Tianyu Kong**, Mitchell Luskin, Alexander B. Watson
   arXiv preprint. arXiv:2502.08120
3. **Modeling of electronic dynamics in twisted bilayer graphene**
   Tianyu Kong, Diyi Liu, Mitchell Luskin, Alexander B. Watson
   SIAM Journal on Applied Mathematics. 84, 1011 (2024).
4. **Bistritzer-MacDonald dynamics in twisted bilayer graphene**
   Alexander B. Watson, **Tianyu Kong**, Allan H. MacDonald, Mitchell Luskin
   Journal of Mathematical Physics 64.3 (2023). Editor's Choice

Research Experience

Graduate Research Assistant, UMN — *06.2021 – Present*
Advisor: Mitchell Luskin and Alexander B. Watson

- Studied mathematical modeling and numerical analysis for incommensurate 2D materials, specifically twisted bilayer graphene
- Accurately computed the band structure and single electron dynamics in tight binding models of such systems
- Used multiple-scale analysis to identify a regime where the discrete aperiodic model can be reduced to a periodic continuum model

- Derived a domain truncation scheme to approximate the electron dynamics in an infinite system.
- Implemented Python code to numerically verify results

Student Assistant, Lawrence Berkeley National Laboratory *05.2024 – 08.2024*
Advisor: Chao Yang and Lin Lin *05.2023 – 08.2023*

- Studied exotic quantum phases in magic angle twisted bilayer graphene using interacting models that accounts for electron-electron correlation
- Implemented Hartree-Fock (HF) and Coupled Clusters (CC) methods with Python-based Simulation of Chemistry Framework (PySCF)
- Developed an accurate continuum model to compute the electron dispersion with structural relaxation, and computed long range Coulomb interaction between electrons

Undergraduate Research Assistant, UChicago *04.2020 – 05.2021*
Advisor: Mary Silber

- Studied different types of Minimum Action Methods to determine the transition of states in a dynamical system subject to random perturbation
- Implemented MATLAB code and used steepest descent and quasi-Newton methods to calculate the Minimum Action Path in Lorenz systems

Teaching Experience

Teaching assistant, School of Mathematics, UMN *09.2021 – 05.2024*
Honors Calculus
Calculus for College of Science and Engineering

- Held 2 workshop sessions per week for 3 hours total, lectured and organized group discussions on challenging concepts and problems, and responded to students' questions in an engaging manner.
- Graded weekly homework, quizzes and final exams promptly, and provided valuable feedback.

Grader, School of Mathematics, UMN *09.2022 – 05.2024*
Mathematical Modeling and Applied Mathematics
Functional Analysis

- Promptly graded homework and exams for graduate level courses.

Grader, Department of Mathematics, UChicago *09.2018 – 05.2020*
Mathematical Methods for Physical Sciences
Honors Calculus

- Promptly graded weekly homework and exams for $\sim$ 30 students.

Talks and Seminar Presentations

Modeling Electron Interactions in Twisted Bilayer Graphene *03.2025*
UMN Quantum Math Seminar, Minneapolis MN

"Magic" in Moiré Materials – An Applied Mathematics Perspective *03.2025*
UMN Seminar of Mathematical Applications & Computations, Minneapolis MN

Modeling of Electronic Dynamics in Twisted Bilayer Graphene *07.2024*
International workshop on 2D and moiré materials, Roscoff France

Modeling of Electronic Dynamics in Twisted Bilayer Graphene *05.2024*
SIAM Conference on Materials Science (MS24), Pittsburgh, PA

Modeling of Electronic Dynamics in Twisted Bilayer Graphene *03.2024*
Brin Mathematics Research Center Workshop, College Park, MD

Poster Presentations

Interacting Twisted Bilayer Graphene with Systematic Modeling of Structural Relaxation *05.2025*

Simons Foundation Moiré Materials Magic Workshop, New York, NY

A Comparison of Minimum Action Methods for Computing Noise-induced Transitions of the Lorenz System *05.2021*

SIAM Conference on Applications of Dynamical Systems (DS21), Remote

Professional Affiliations

Webmaster, SIAM student chapter, UMN *09.2024 – Present*

- Managed website and documents for SIAM student chapter at UMN
- Co-organized annual student Integration Bee contest

Technologies

Python, MATLAB, Julia, R, MySQL